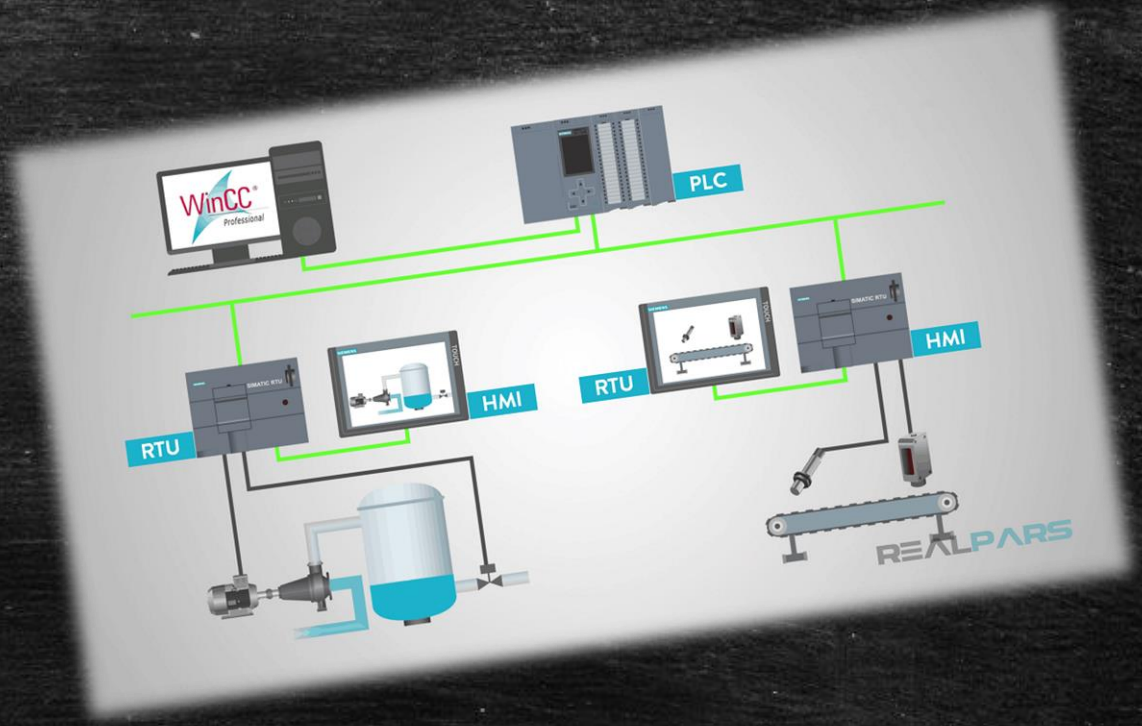


SCADA

GNB

Content

- Introduction
- Application Areas of SCADA
- Major Elements of SCADA systems
- Comparison of SCADA, DCS and PLC
- Considerations and benefits of SCADA system.

Introduction

- SCADA is an acronym of Supervisory Control and Data Acquisition.
- It is a technology that enables the user to collect data from one or more distant facilities and to send limited control instructions to those facilities.
- It makes unnecessary for an operator to stay or frequently visit the plant where remote facilities are operating normally.
- SCADA refers to the combination of telemetry and data acquisition.
- SCADA allows an operator in a location central to a widely distributed process such as oil or gas field, pipeline system, irrigation system or hydroelectric generating complex to make set point changes on distant process controllers, to open or close valves or switches, to monitor alarms and to gather measurement information.

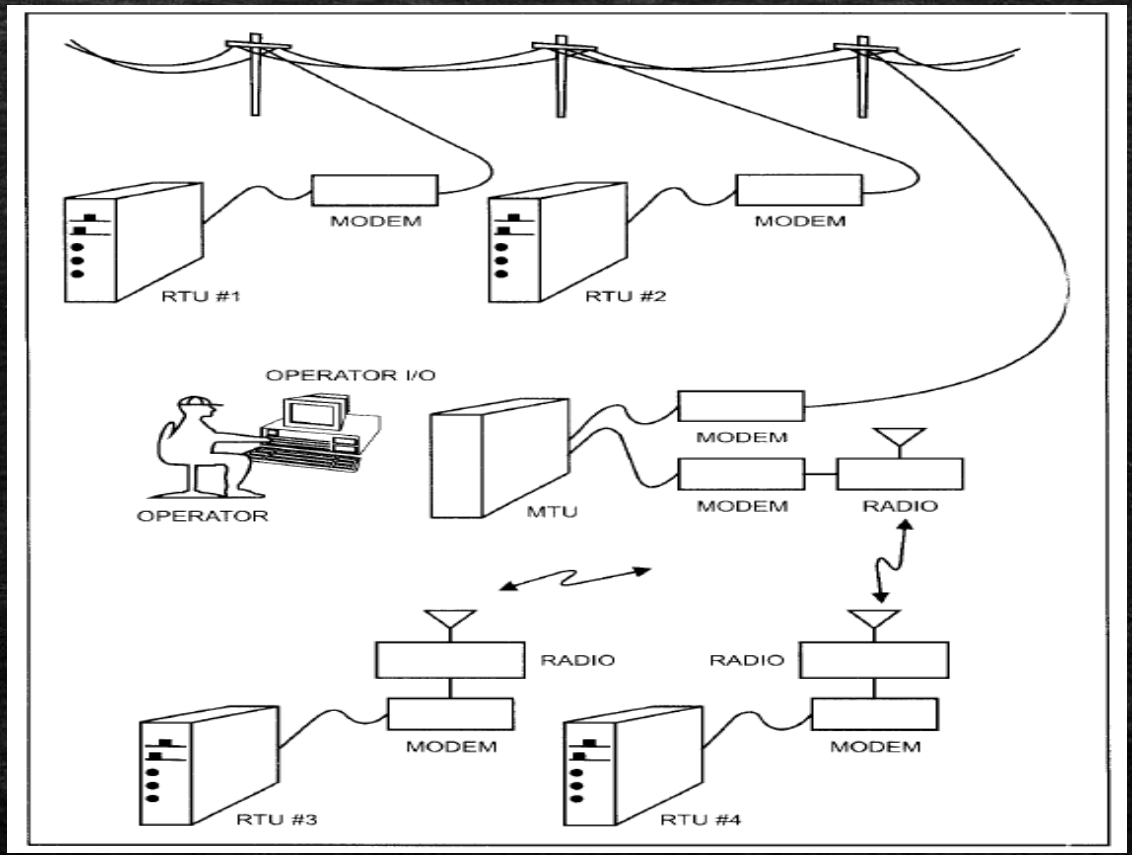
Applications

- SCADA Technology is applicable to the process that are spread over large areas, relatively simple to control and monitor and require frequent, regular or immediate intervention. They are applicable for:
- Groups of small hydroelectric generating stations that are turned ON/OFF in response of the customer demand that are usually located in remote locations. They can be controlled by opening and closing of valves of turbines, they must be monitored continuously and they need to respond relatively quickly to demands on the electric power grid.
- Oil or gas production facilities – including wells, gathering systems, fluid measurement equipment and pumps are usually spread over large areas require relatively simple control like ON/OFF, meter information.
- Electric Transmission, Pipeline of gas, oil, chemicals or water.

Elements of SCADA

Elements of SCADA

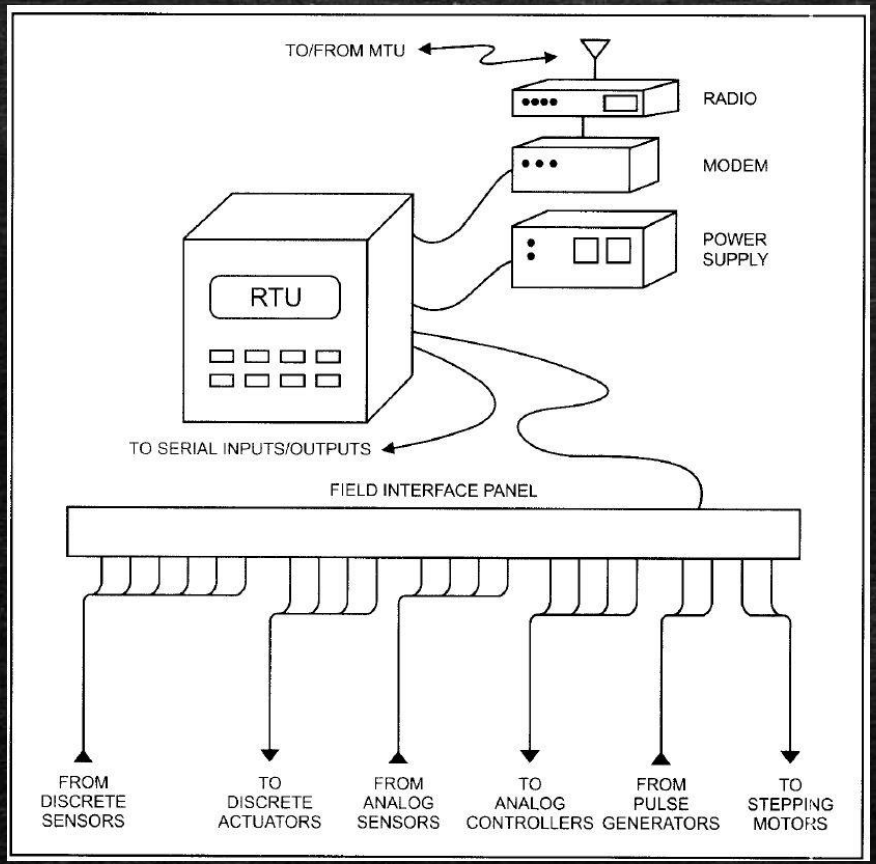
- Remote Terminal Unit
- Master Terminal Unit
- Operator I/O Interface
- Modem
- Communication Mode



Elements of SCADA

- At the center is the operator, who can access the system by means of an operator interface device called as operator console. It consists of VDU that displays RT data about the process and suitable input devices.
- Master Terminal Unit is the System Controller. It is also called as Host computer or server. It is based on computer. It can monitor and control the field even when the operator is not present. It is done by inbuilt scheduler that can be programmed to repeat instructions at set intervals. MTUs must communicate with RTU that are located away from the central location.
- Communication can be achieved by two ways: landline, which takes in form of optical fiber or electrical cable. Radio and Telephone Lines. Both uses Modem
- RTU communicates with MTU and must be capable of understanding messages. It is connected to field devices by wired technique.

Remote Terminal Unit

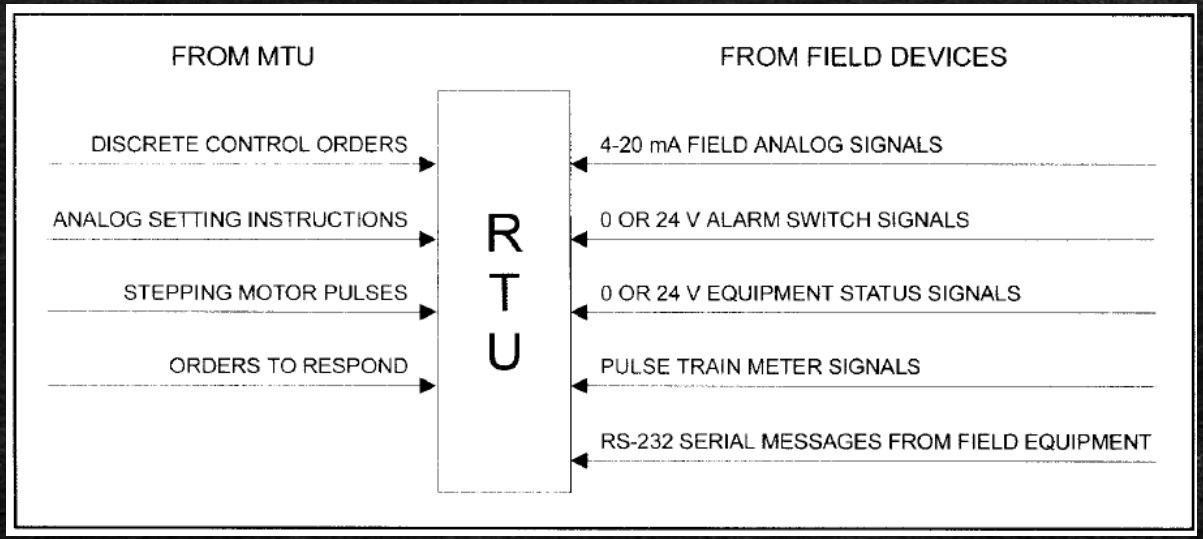


Remote Terminal Unit

- RTU gathers information from the field about analog values, alarm and status points and metered amounts.
- It keeps this information available in memory until the MTU asks for it.
- It then codes and transmits the information to the MTU.
- In addition, when MTU instructs, the RTU opens and closes valves, turns switches ON/OFF, outputs analog signals that may represent set points and output pulse trains to move stepping motors.
- They have the ability to receive and send messages to field devices in serial format, usually 232.

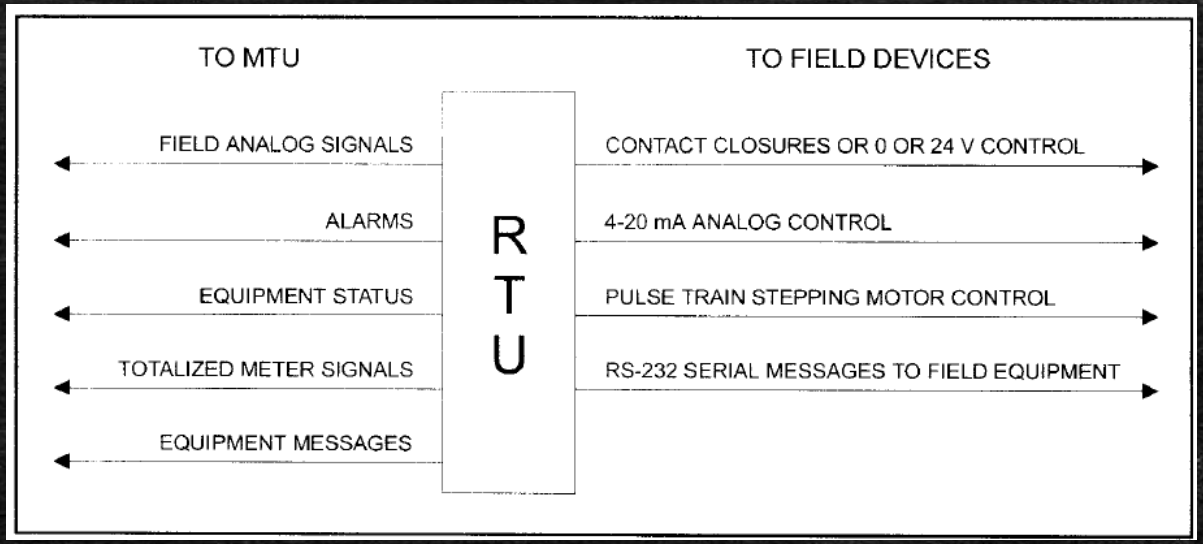
Remote Terminal Unit Connections

The signals that come into the RTU

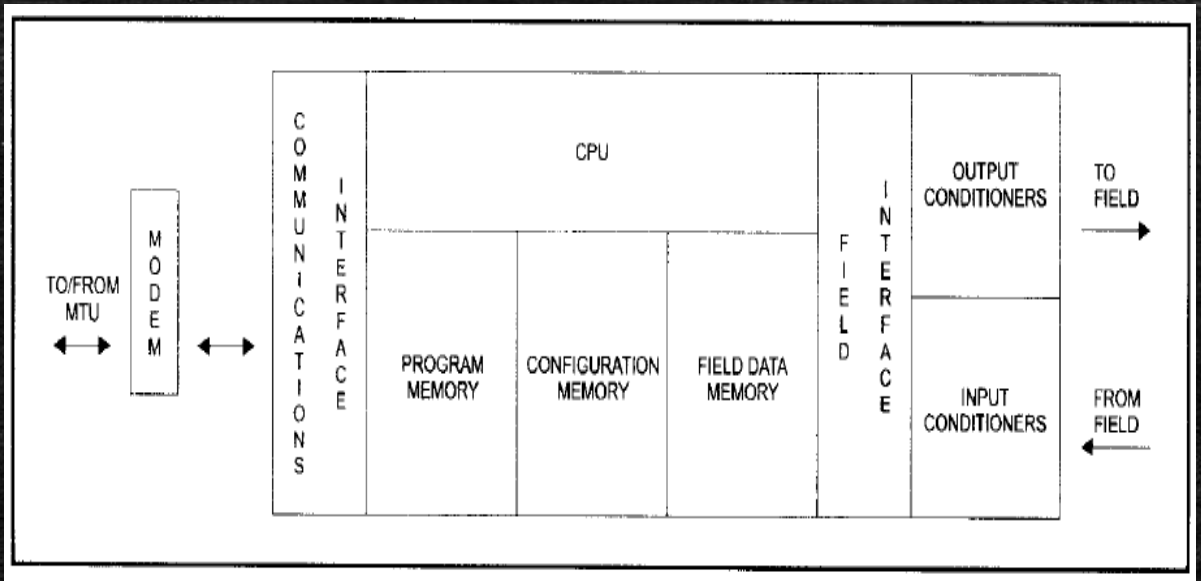


Remote Terminal Unit Connections

The signals that leaves the RTU

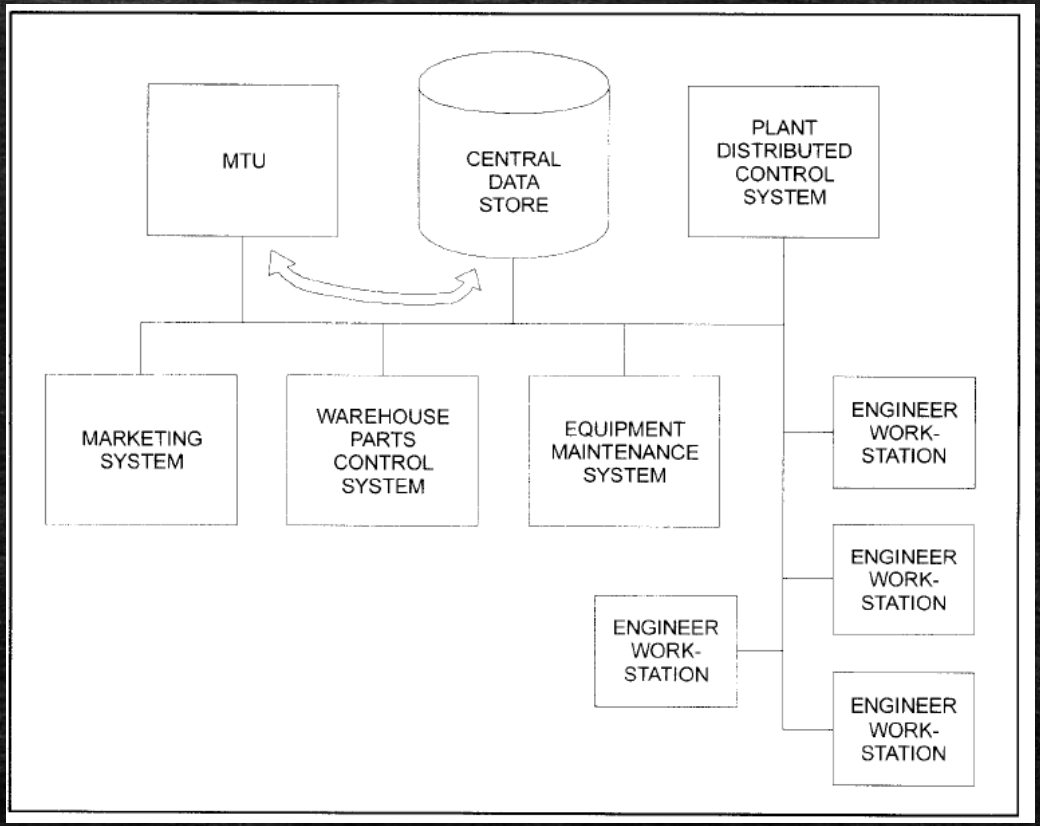


Remote Terminal Unit



Master Terminal Unit

MTU is the device that issues all commands, gathers all the data, stores some information, passes other information onto associated systems, interfaces with the people who operates the process.



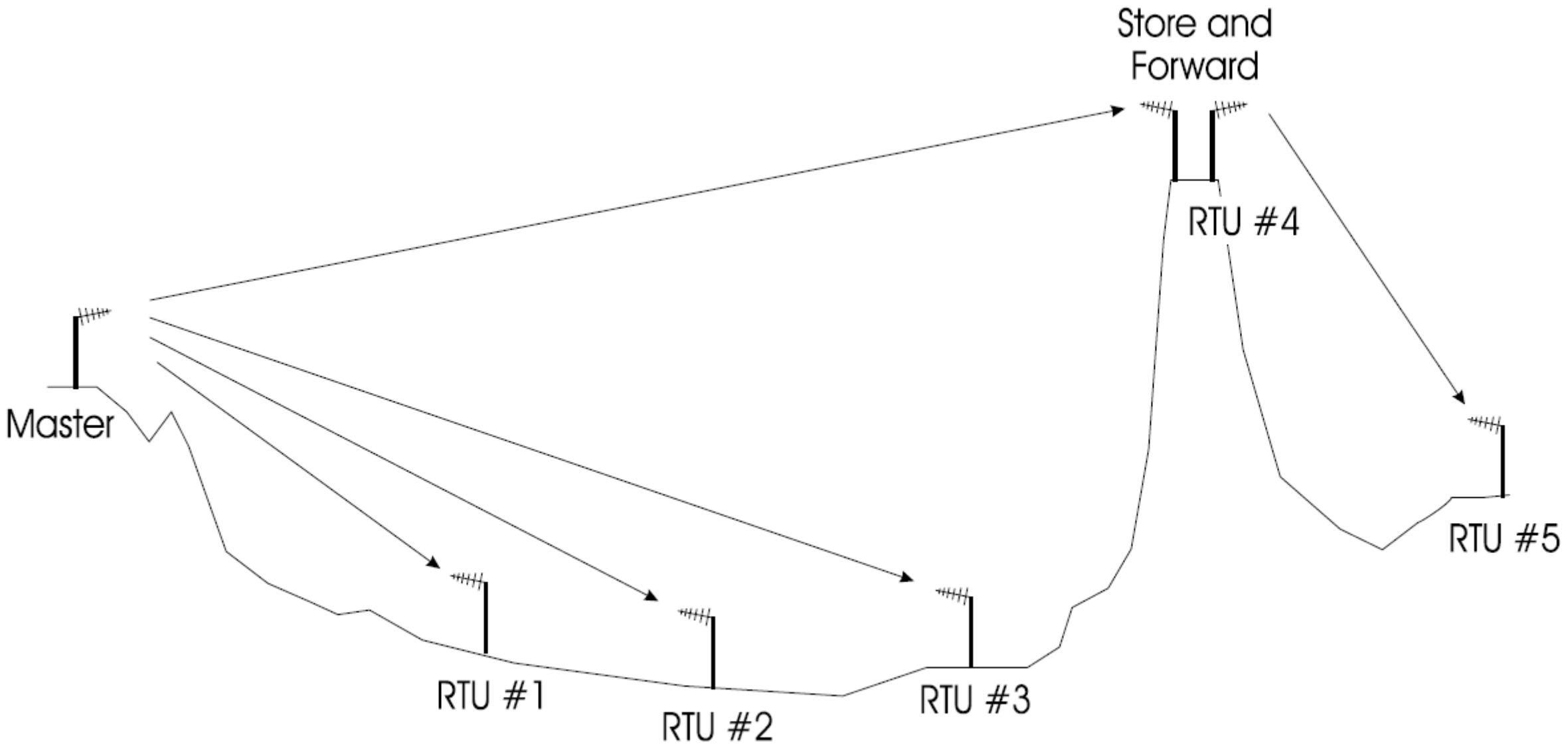
Master Terminal Unit

- It has two main functions: to obtain field data from RTU periodically & to control remote devices through operator station.
- It has three components: Operating System software, System SCADA software and Scada Application Software.
- System SCADA software refers to the software consisting of four main modules: data acquisition, control, archiving and man machine interface.

Master Terminal Unit - Functions

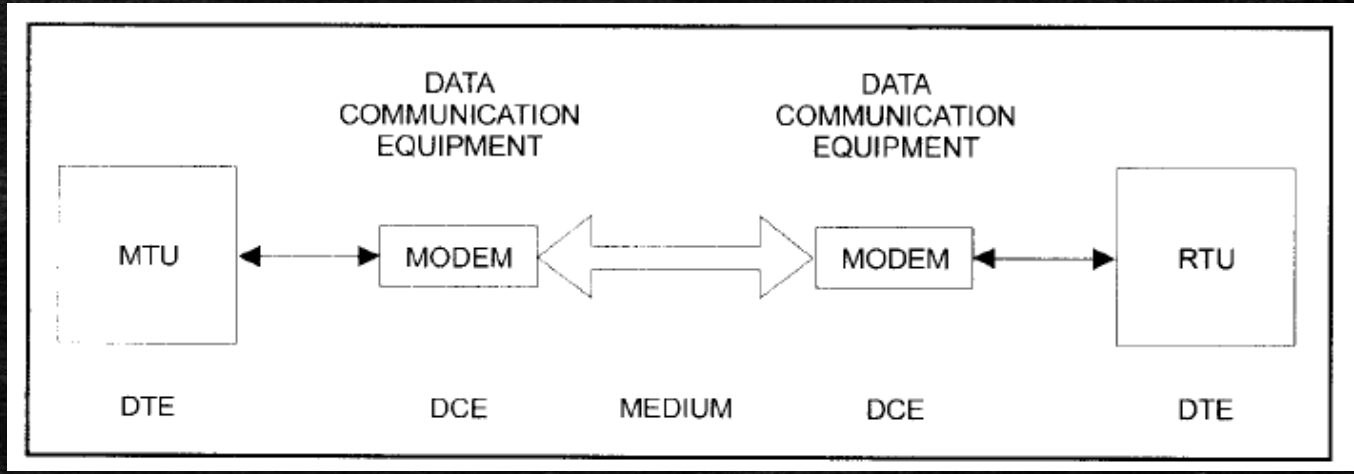
- **Establishment of communications**
  1. Configure each RTU
  2. Initialize each RTU with input/output parameters
  3. Download control and data acquisition programs to the RTU
- **Operation of the communications link**
  1. If a master slave arrangement, poll each RTU for data and write to RTU
  2. Log alarms and events to hard disk (and operator display if necessary)
  3. Link inputs and outputs at different RTUs automatically
- **Diagnostics**
  1. Provide accurate diagnostic information on failure of RTU and possible problems
  2. Predict potential problems such as data overloads

Master Terminal Unit



Communications

- Communication is the movement of data or intelligence from one location to another.
- First , communication path must exist, some medium must be selected over which the data will travel.
- Second, equipment must exist at the sending end of the communication path to condition the data and to put into a form that can be sent over communication medium.
- Third, equipment must exist at the receiving end of the path to extract the message from the medium and understand its meaning.



Communications – Telephonic cable or Radio

- In the past the choice of communication media was determined by two things: data rate and cost.
- Now it is more focused on security of data.
- Lines leased from the telephone company are one possibility for communications medium. The cost is likely high but for those industries that require high volume, high speed data, these choices do exist.
- Data rate is about 300 to 4800 bps range.
- UHF radios have been developed specially for scada applications. The offer low cost, high reliability and flexibility.

Considerations of SCADA

- Overall control requirements
- Sequence logic
- Analog loop control
- Ratio and number of analog to digital points
- Speed of control and data acquisition
- Master/operator control stations
- Type of displays required
- Historical archiving requirements
- System consideration
- Reliability/availability
- Speed of communications/update time/system scan rates
- System redundancy
- Expansion capability
- Application software and modelling

Benefits of SCADA

- Improved operation of the plant or process resulting in savings due to optimization of the system
- Increased productivity of the personnel
- Improved safety of the system due to better information and improved control
- Protection of the plant equipment
- Safeguarding the environment from a failure of the system
- Improved energy savings due to optimization of the plant
- Improved and quicker receipt of data so that clients can be invoiced more quickly and accurately
- Government regulations for safety and metering of gas (for royalties & tax etc)

Comparison between PLC, DCS and SCADA

	PLC	DCS	SCADA
Functions	Single process control	Entire plant control	Provides communication between equipment in a plant
Speed	Faster than DCS & SCADA	Faster than SCADA	Slower than DCS
Networking	Engineered networking as per requirements	All the components are tightly integrated into the network	All the components are tightly integrated into the network
Communication	Doesn't have a wide range of communication	Wide range of communication	Wide range of communication
Margin for error	Only one device controls the process so there is a chance for error	There are many controllers so less margin for error	Less margin for error
Pricing	Low cost	High initial cost	High initial cost
Major brands	Allen Bradley, Texas Instruments, Omron, etc.	Honeywell, Yokogawa, Siemens, etc.	Allen Bradley, Siemens, MODICON etc.
Monitoring range	Short distance (single process)	Entire plant	Long-distance monitoring
System composition	Hardware	Hardware, Visualizing software, and historian	No hardware, Data collection, and monitoring
Flexibility	Highly flexible	Due to the complex architecture, it is not flexible as PLC	Due to the complex architecture, it is not flexible as PLC
Process changes	It can handle process changes	Due to the complexity, it won't be able to handle the process change	Due to the complexity, it won't be able to handle the process change
Applications	Used for dedicated applications	Used for complex applications	Used in big industries (Remote monitoring)

END